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Roll No.....

EE-8302 (GS)
B.E. VIII Semester Examination, June 2020
Grading System (GS)
Process Control
(Elective - III)
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.

1. What is a Ratio Control System? Describe such a control system with a specific process.
2. a) Explain the application of PLC and its uses in process control.
b) What is supervisory control? Explain its features in process control.
3. What are the principle characteristics of the first-order processes and what causes the appearance of a purely capacitive process?
4. Define and explain effect of control parameters.
 - i) Pure lag
 - ii) Time constant of any process.
5. Describe the following process characteristics:
 - i) Process equation
 - ii) Process load
 - iii) Process lag
 - iv) Self regulation
6. Mention any two drawbacks of the derivative control action.

OR

Give the optimum controller settings for the model
$$G(s) = \frac{e^{-0.5s}}{4s + 1}$$
obtained by reaction curve method.
7. Draw the block diagram of data loggers and explain it.
8. Write short notes on: (any two)
 - a) Heat exchangers
 - b) Volume boosters
 - c) Safety valves
